

Course Name: Python Programming
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Practice Questions

Section A - MCQ

Q1. Which of the following is a valid Python variable name?

- a. 1var
- b. Var_1
- c. Var-1
- d. var 1

Q2. Which function returns the data type of a variable?

- a. id()
- b. type()
- c. len()
- d. input()

Q3. Which statement does nothing but avoid error?

- a. pass
- b. break
- c. continue
- d. exit

Q4. Which method removes the last list element?

- a. delete()
- b. pop()
- c. remove()
- d. clear()

Q5. How are dictionary values accessed?

- a. index
- b. keys
- c. key name
- d. position

Q6. Object Identity Trap

x = 257

y = 257

print(x is y, x == y)

Output: _____

Q7. String Interning & Slicing

```
s1 = "py" + "thon"  
s2 = "python"  
print(s1 is s2, s1[::2])
```

Output: _____

Q8. Loop–else Construct

```
for i in range(3):  
    if i == 5:  
        break  
else:  
    print("Loop completed")
```

Output: _____

Q9. Mutable Object Multiplication

```
x = [1, 2]  
y = x * 2  
y[0] = 99  
print(x, y)
```

Output: _____

Q10. Short-Circuit Evaluation

```
a = 0  
b = 5  
print(a and b, a or b)
```

Output: _____

Q11. Tuple vs List Mutability

```
t = (10, 20, [30, 40])  
t[2] += [50]  
print(t)
```

- ☐ Executes successfully
- ☐ Raises exception

Answer (output or exception name with explanation): _____

Q12. Dictionary Key Overwriting

```
d = {True: "yes", 1: "one", 1.0: "float"}  
print(d)
```

Output: _____

Q13. Control Flow with continue

```
i = 0
while i < 5:
    i += 1
    if i % 2 == 0:
        continue
    print(i, end=" ")
```

Output: _____

Q14. String Method Logic

Fill in **ONE** method so the output is True.

```
s = "Python3"
print(s._____.())
```

Q15. id() Behaviour

```
a = [1, 2, 3]
b = a[:]
print(id(a) == id(b))
```

Output: _____

Section B – Coding Question**Q1. Remove Duplicates from List (Without Using set)**

Problem: Given a list, remove duplicates while maintaining order.

Example:

Input: [1, 2, 2, 3, 4, 1, 5]

Output: [1, 2, 3, 4, 5]

Test Cases:

Input: [1,1,1] -> Output: [1]

Input: [] -> Output: []

Input: [5,4,3,2,1] -> Output: [5,4,3,2,1]

Q2. Rotate List by k Positions

Problem: Rotate the list to the right by k steps.

Example:

Input: nums = [1,2,3,4,5], k = 2

Output: [4,5,1,2,3]

Test Cases:

Input: [1,2,3], k=1 -> Output: [3,1,2]

Input: [1], k=10 -> Output: [1]

Input: [], k=3 -> Output: []

Q3. Flatten a Nested List

Problem: Flatten a nested list of any depth.

Example:

Input: [1, [2, [3, 4]], 5]

Output: [1,2,3,4,5]

Test Cases:

Input: [] -> Output: []

Input: [[1,2],[3,4]] -> Output: [1,2,3,4]

Q4. Count Frequency of Elements

Problem: Return dictionary with frequency count.

Example:

Input: [1,2,2,3,1,1]

Output: {1:3, 2:2, 3:1}

Test Cases:

Input: [] -> Output: {}

Input: ['a','b','a'] -> Output: {'a':2,'b':1}

Q5. Merge Two Dictionaries

Problem: If the key exists in both dictionaries, sum their values.

Example:

d1 = {'a':10,'b':20}

d2 = {'b':5,'c':15}

Output: {'a':10,'b':25,'c':15}

Test Cases:

Input: {}, {} -> Output: {}

Input: {'x':1}, {'x':2} -> Output: {'x':3}

Q6. Find Key with Maximum Value

Problem: Return the key that has the highest value.

Example:

Input: {'a':10,'b':50,'c':20}

Output: 'b'

Test Cases:

Input: {'x':1} -> Output: 'x'

Input: {} -> Output: None

Q7. Convert a string where words are separated by spaces to one where words are separated by underscores.

Q8. Check if a string starts with a specific word or phrase.

Q9. Find the longest word in a string.

Q10. Extract all digits from a string.

Q11. Replace multiple spaces in a string with a single space.

Q12. Write a Python program that accepts a string as input, then

(a) Displays the first half of the string.

(b) String in reverse order

(Do not use built-in reverse functions)

Q13. Write a Python program to:

(a) Create a tuple of student marks.

(b) Find the highest and lowest marks.

(c) Display the result.

Q14. Write a Python program to:

(a) Store employee name as key and salary as value.

(b) Display employee names who are earning more than ₹50,000.

Q15. Write a Python program to:

(a) Store student roll number as key

(b) Store name and marks as values in a dictionary

(c) Display students who have scored **more than 75 marks**

Q16. Write a Python program that:

(a) Accepts a **single-line string input**.

(b) Creates a **dictionary** where:

(i) Keys are **unique characters** (excluding spaces).

(ii) Values are **lists of all index positions** where that character occurs.

The program must be **case-sensitive**.

Example Input

aAb a

Expected Output

```
{
'a': [0, 4],
'A': [1],
'b': [2]
}
```

Constraints

- Use range() and string indexing.
- Use dictionary methods only.
- No external modules.